# Modelling and Simulation of Nonlinear and Ecological Natural Systems

Yongjiang

## Abstract

This report studies three natural-system models using the modelling language introduced in the module: high-order differential equations are converted into coupled first-order systems, numerical methods are used to approximate initial-value problems, and agent-based simulations are contrasted with equation-based models. The Duffing oscillator is simulated with Runge–Kutta methods and analysed through phase portraits, discretisation-step convergence and sensitivity to initial conditions. A prey–predator agent-based model is then used to study emergent stochastic oscillations and parameter sensitivity. Finally, a harvested Lotka–Volterra model is analysed numerically and placed in the context of classical and extended prey–predator models.

## 1 Duffing Oscillator

The forced Duffing oscillator in the assignment is

$$\frac{d^2x}{dt^2} = \gamma\cos(\omega t) - \delta\frac{dx}{dt} - \alpha x - \beta x^3, \tag{1}$$

with $\gamma = 0.29$, $\delta = 0.3$, $\alpha = -1$, $\beta = 1$ and $\omega = 1.2$. Following the lecture treatment of higher-order systems, define

$$x_1(t) = x(t), \qquad x_2(t) = \frac{dx}{dt}. \tag{2}$$

The model becomes the first-order, vectorial system

$$\frac{dx_1}{dt} = x_2, \qquad \frac{dx_2}{dt} = \gamma\cos(\omega t) - \delta x_2 - \alpha x_1 - \beta x_1^3. \tag{3}$$

For the given parameters this is

$$\frac{dx_2}{dt} = 0.29\cos(1.2t) - 0.3x_2 + x_1 - x_1^3. \tag{4}$$

This is suitable for numerical simulation because it has the standard module form $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$ with $\mathbf{x} = (x_1, x_2)^T$.

I used the fourth-order Runge–Kutta method (RK4). This choice is conservative for a nonlinear forced oscillator because, for the same discretisation step, RK methods provide more accurate approximations than Euler's method. Figure 1 shows solutions for different initial conditions. The time series do not settle to a fixed equilibrium. Instead, the solution is attracted to bounded oscillatory motion whose transient depends on the initial condition. The phase portrait reports $x(t)$ against $v(t) = dx/dt$, as required by the assignment, and shows that the trajectory is bounded but not a simple one-dimensional equilibrium approach.

The discretisation step was selected by comparing RK4 simulations against a fine reference solution with $\Delta t = 0.001$. Figure 2 shows both trajectory-level comparisons and the root-mean-square error. The errors decreased from $1.28 \times 10^{-1}$ at $\Delta t = 0.2$ to $1.08 \times 10^{-6}$ at $\Delta t = 0.01$, and to $7.05 \times 10^{-8}$ at $\Delta t = 0.005$. I therefore used $\Delta t = 0.01$ in the simulations: it is already visually indistinguishable from the reference at the scale of the figures while keeping the computational cost moderate.

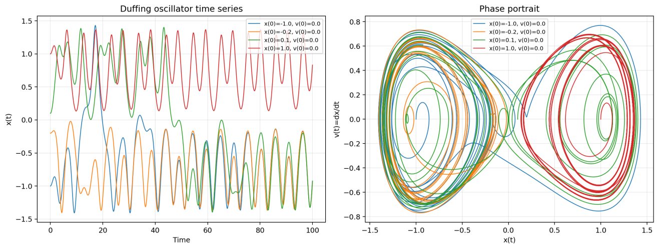



**Figure 1:** Duffing oscillator with $\gamma = 0.29$ for different initial conditions. Left: trends of $x(t)$ as time varies. Right: phase portrait in the $(x, v)$ plane.

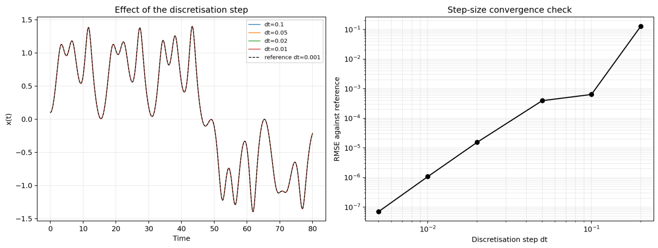


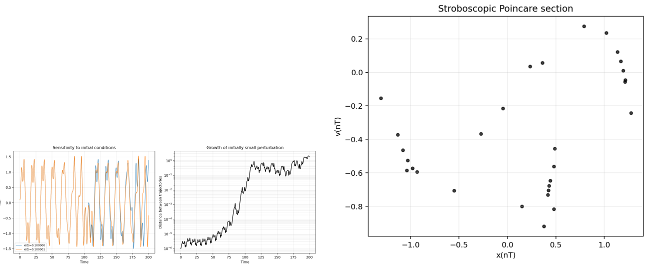

**Figure 2:** Selection of the discretisation step. Several RK4 solutions are compared with a reference solution, and the error decreases systematically as $\Delta t$ is refined.

The system is non-autonomous because the right-hand side explicitly depends on time through the forcing term $\gamma\cos(\omega t)$. In the lecture terminology, an autonomous system would have the form $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ without explicit $t$-dependence. The Duffing system could be embedded into a higher-dimensional autonomous system by introducing the forcing phase as an additional state variable, but in the form given in the assignment it is non-autonomous.

For Exercise 2, I set $\gamma = 0.5$ and used RK4. A classic feature of chaotic systems is sensitivity to initial conditions. Figure 3 compares two trajectories with initial separation $10^{-6}$: $(x(0), v(0)) = (0.100000, 0)$ and $(0.100001, 0)$. The separation grew to a maximum of approximately 2.15, which is not compatible with a merely stable periodic response to small perturbations. As a second approach, I used a stroboscopic Poincare section sampled once per forcing period $T = 2\pi/\omega$. The points do not collapse to a single fixed point or a short periodic orbit, supporting the interpretation that the forced system is in a chaotic regime.

## 2 Agent-Based Prey–Predator Model

The agent-based model was implemented using the lecture ecolab structure: an environment stores grass, and rabbit and fox agents inherit from a common agent class. Each iteration applies local rules for movement, eating, breeding and death. This follows the lecture definition of agent-based modelling as a programmatic approach based on local interactions and




**(a)** Initial-condition sensitivity. **(b)** Poincare section.

**Figure 3:** Evidence for chaotic behaviour when $\gamma = 0.5$. The response is supported both by sensitivity to initial conditions and by a stroboscopic phase-space view.

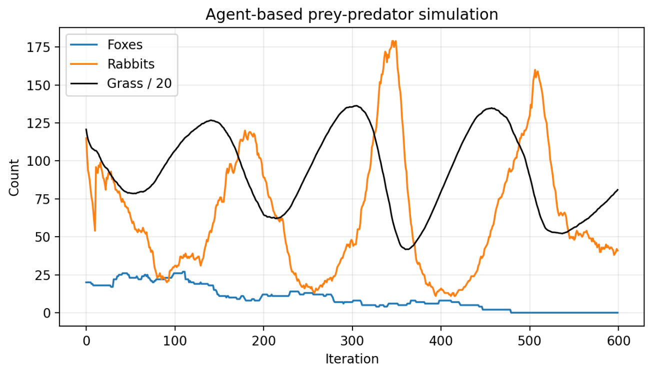



**Figure 4:** Agent-based prey–predator simulation using local rabbit, fox and grass rules. Grass is divided by 20 for display on the same axis.

emergent behaviour, rather than starting directly from global differential equations.

Figure 4 reports a representative simulation. The populations display irregular oscillatory behaviour: rabbits grow when grass is available, foxes subsequently increase when rabbits are abundant, and rabbit abundance then falls under predation pressure. Grass is scaled by a factor of 20 only to plot it on the same axis. This reproduces the typical prey–predator feedback qualitatively, but differs from equation-based Lotka–Volterra modelling in several ways. First, the ABM is stochastic, so repeated runs differ even with identical parameters. Secondly, interactions are local: a fox can only catch nearby rabbits, whereas ODE models use mean-field global interaction terms. Thirdly, agents are discrete individuals with age, food, speed and breeding constraints; these mechanisms are not represented in the basic ODE. Finally, extinction occurs naturally because integer populations can reach zero, while continuous ODE solutions may approach small positive values without representing individual loss.

To examine parameter sensitivity, I varied the fox speed $\alpha$ from 2 to 5 and performed eight independent simulations for each value. The variables of interest were rabbit extinction probability, final rabbit count, and maximum/minimum rabbit count over a 600-iteration simulation. Figure 5 shows a clear trend: increasing fox speed increases predation efficiency and substantially raises extinction risk. The estimated rabbit extinction probability increased from 0 at speeds 2 and 3 to 0.375 at speed 4 and 0.875 at speed 5. The final rabbit count also decreased on average. The standard deviations are large, which is expected in a stochastic ABM and is why reporting averages alone would be misleading.

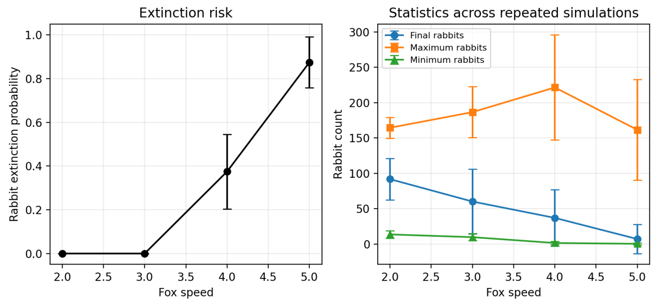



**Figure 5:** Sensitivity analysis for the agent-based model. Error bars report variability across repeated independent simulations.

# 3 Prey–Predator Equations with Harvesting

For $h = 0$, the assignment system becomes the standard Lotka–Volterra model,

$$\dot{x} = x(\alpha - \beta y), \qquad \dot{y} = y(\delta x - \gamma). \tag{5}$$

I used $\alpha = 1.0$, $\beta = 0.1$, $\gamma = 1.5$ and $\delta = 0.075$. With these parameters, the non-zero equilibrium is

$$x^* = \frac{\gamma}{\delta} = 20, \qquad y^* = \frac{\alpha}{\beta} = 10. \tag{6}$$

The standard model produces closed prey–predator oscillations around this coexistence point: prey first increase when predators are scarce, predators then increase after prey become abundant, and the subsequent predation reduces prey numbers.

For $h > 0$, harvesting subtracts a constant amount from the predator equation:

$$\dot{y} = y(\delta x - \gamma) - h. \tag{7}$$

Figure 6 shows simulations for increasing $h$. Small harvesting values reduce predator minima and distort the phase portrait while retaining non-negative populations over the simulated interval. Larger harvesting values drive the predator population to the boundary; for $h = 0.4$ and $h = 0.8$ the numerical solution becomes biologically implausible at approximately $t = 34.15$ and $t = 21.74$, respectively, because predator abundance crosses below zero. The biological interpretation is that the ODE is meaningful only while $x(t) \geq 0$ and $y(t) \geq 0$. Once a population becomes negative, the mathematical solution no longer represents a prey–predator system, and the model should be modified, for example by imposing extinction at zero or by using a harvesting term that vanishes when predators are absent.

The classical prey–predator formulation originates from Lotka's physical-biology work and Volterra's mathematical treatment of species fluctuations [1, 2]. Its central modelling assumptions are mass-action encounters, homogeneous mixing, unlimited exponential prey growth in the absence of predators, and predator growth proportional to prey consumption. These assumptions make the model analytically transparent but ecologically idealised. Later variants relax different assumptions. Holling introduced functional responses, replacing the linear predation term with saturating consumption caused by handling time and predator limitations [3, 4]. Rosenzweig and MacArthur developed graphical stability conditions and models with prey self-limitation, making resource limitation explicit and connecting predator–prey dynamics to stability theory [5]. Rosenzweig's paradox of enrichment further showed

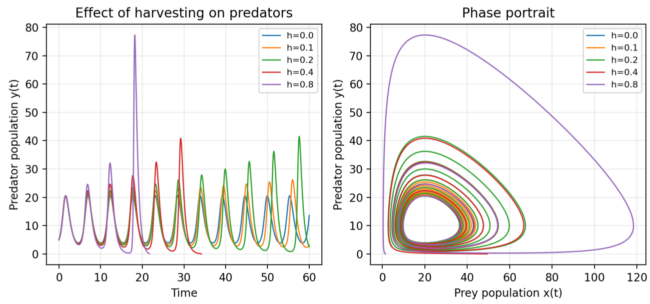



**Figure 6:** Lotka–Volterra predator harvesting. Increasing $h$ reduces predator abundance and can push the solution outside the biologically meaningful non-negative region.

that changing resource availability can destabilise exploitative ecosystems [6]. In the language of this module, these variants encode different modelling assumptions: whether interactions are mean-field or local, whether growth is unlimited or density-limited, whether predation is linear or saturating, and whether human intervention such as harvesting is externally imposed.

# 4 Conclusion

Across the three questions, the same course methodology is used: define the model variables, convert the equations to a first-order system when needed, select a numerical method, verify the discretisation step, and interpret figures in relation to the model assumptions. The Duffing oscillator illustrates nonlinear non-autonomous dynamics and chaos; the agent-based model shows stochastic emergent prey–predator oscillations from local rules; and the harvested Lotka–Volterra model demonstrates how a simple equation-based system can lose biological plausibility when a modelling assumption is pushed too far.

# References

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[3] C. S. Holling, "The components of predation as revealed by a study of small-mammal predation of the European pine sawfly," *The Canadian Entomologist*, vol. 91, pp. 293–320, 1959.

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